European Commission

Mutual Learning Exercise on National policies for Artificial Intelligence in science

Final report

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Report



Research and Innovation

Mutual Learning Exercise on National Policies for AI in Science. Final Report.

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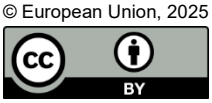
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EUROPEAN COMMISSION

Mutual Learning Exercise on National policies for Artificial Intelligence in science

Final Report

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2025

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List of abbreviations

AI	Artificial Intelligence
DG RTD	Directorate-General for Research and Innovation
EOSC	European Open Science Cloud
ERA	European Research Area
EU	European Union
FAIR	Findable, Accessible, Interoperable, and Reusable (data principles)
GDPR	General Data Protection Regulation
HPC	High-Performance Computing
LLM	Large Language Model
ML	Machine Learning
MLE	Mutual Learning Exercise
OECD	Organisation for Economic Co-operation and Development
PSF	Policy Support Facility
R&D	Research and Development
R&I	Research and Innovation
RAISE	Resource for AI Science in Europe
SSH	Social Sciences and Humanities
STEM	Science, Technology, Engineering, and Mathematics

Executive summary

AI in science is the application of AI to science. It covers everything from individual scientists using traditional AI or machine learning (ML) techniques, deep learning, large language models (LLMs) and related foundation models, through to the development of AI technology tailored for scientific needs in various disciplines, and to fully autonomous AI systems - "AI scientists" - that can conduct scientific research on their own.

AI promises to accelerate the pace of scientific discovery and revolutionise how we uncover and use new knowledge, driving breakthroughs across disciplines. Yet this is just the beginning: future AI systems will become ever more powerful and accessible.

AI in science holds a prominent place in the European Union's political agenda. In the political guidelines of the European Commission (2024-2029), President Ursula von der Leyen underscores the European ambition to "*focus our efforts on becoming a global leader in AI innovation*". The political mandate of the Commissioner for Startups, Research and Innovation includes the **development of a dedicated AI in Science Strategy and the set-up of an AI Research Council**. This Council will be implemented as a **Resource for AI Science in Europe (RAISE)**, as announced in the AI Continent Action Plan adopted by the European Commission on 9 April 2025. It will pool resources at EU and national level to facilitate faster advances of AI in and through science in Europe. As the current report is being compiled, the European Commission is developing these two strategic initiatives and the insights from the Mutual Learning Exercise (MLE) country visits are expected to inform this work.

This MLE, initiated under the Horizon Europe Policy Support Facility, brought together 11 countries to exchange policy experiences and identify key enablers for the effective integration of AI into scientific research. The process focused on three themes:

1. Framework conditions and funding for AI in science.
2. Infrastructures for AI in science.
3. Talent for AI in science.

A set of policy recommendations for national policy makers and research organisations were developed during the MLE. The recommendations address key challenges identified in the MLE country visits such as limited access to high-performance computing (HPC) and data infrastructure, fragmented and underused scientific data, and persistent talent gaps. The recommendations can be further tailored to countries' stage of AI maturity in science (emerging, expanding, advanced).

While some policy actions may apply across multiple governance levels, each recommendation is presented and discussed where it is likely to have the greatest impact. The policy "menu" proposed in this report is not prescriptive; it does not define one-size-fits-all solutions. Rather, it offers a flexible framework and a set of guidelines that countries can adapt to their contexts and priorities.

Policy recommendations		
Topic	**Policymakers**	**Research organisations**
Funding and Framework Conditions	• Develop national AI in science roadmaps and visions aligned with EU strategies. • Design funding portfolios that combine top-down investments and bottom-up experimentation. • Set up and connect to networks of excellence • Fund interdisciplinary collaboration. • Establish long-term funding for risky and interdisciplinary projects on AI in science and promote co-financing with industry. • Allocate resources to both technical and human infrastructure. • Operationalise ethical frameworks. • Set up anticipatory governance processes, coordinate AI policies across departments. • Support communities of practice. • Fund dedicated support structures (e.g. AI advisors) to assist researchers. • Monitor AI impact on science.	• Offer seed funding, fellowships, and protected spaces to support early AI experimentation. • Develop co-financing mechanisms with industry. • Ensure access to AI funding across departments and disciplines. • Align strategies with national/ERA priorities while preserving research autonomy. • Participate in and help build national and international networks of excellence. • Foster communities of practice across institutions. • Implement FAIR data practices and reward transparency, openness and good data stewardship.
Infrastructures	• Support national HPC and data infrastructures, promote participation in EU-level initiatives. • Incentivise private sector contributions to boost AI resource availability.	• Ensure AI-ready infrastructure accessibility, interoperability and efficiency in use. • Promote cross-disciplinary collaboration and standardise data formats, promote FAIR data practices

Policy recommendations		
Topic	**Policymakers**	**Research organisations**
	• Create data governance frameworks. • Invest in national repositories and cloud infrastructures for secure data access. • Support domain-specific and sustainable AI systems. • Establish benchmarks and guidelines for responsible AI use in science. • Promote engagement in national and European networks of excellence.	• Provide support roles and training in data management. • Encourage academia–industry collaboration with Open Science safeguards. • Connect to national and European networks of excellence.
Talent	• Include AI basics in school curricula. • Foster scientific curiosity about the potential of AI in science. • Support grants, fellowships, and subsidies for interdisciplinary AI training. • Support national HPC-related skill development programmes. • Create AI research centres and co-financing models with industry.	• Educate researchers about AI models and risks. • Establish structured career paths and incentives for AI in science. • Reform evaluation criteria to recognise AI training and interdisciplinary research. • Expand AI education across disciplines. • Support interdisciplinary PhD/postdoc programmes. • Support mobility between academia and industry.

Table 1: Policy recommendations for AI in Science for policy makers and research organisations.

AI in science sits at the intersection of different national policy agendas, in particular AI policies and research and innovation (R&I) policies. While no Member State or Associated Country of Horizon Europe has a dedicated AI in science strategy, the majority of AI policies or national R&I strategies of the MLE countries do contain initiatives that address it, albeit in different formats and scopes:

- AI strategies often focus on enabling conditions for AI research and development (R&D) such as infrastructure, ethical frameworks, and skills, but tend to address the broader AI ecosystem rather than specifically targeting the needs of the research sector.
- R&I policies often focus on framework conditions such as research funding (often interdisciplinary), talent attraction, translation and knowledge exchange, and research culture.

Although countries are at different stages in developing and adopting AI in science and national contexts vary widely, **the MLE revealed a shared recognition of AI's transformative potential for science and competitiveness, and of the urgent need for strategic, coordinated policy responses**. Policy frameworks must be responsive to rapid technological change, supporting experimentation while building long-term capability. The MLE identified several success factors for effective AI in science policy:

- **Pairing strategic investments with bottom-up experimentation**: An effective investment agenda combines large-scale, coordinated funding with flexible mechanisms that allow researchers to seize domain-specific opportunities. Support structures must also be adapted to local contexts.
- **Building human and technical infrastructures**: Beyond compute resources and data platforms, AI in science requires sustained investment in human capital (e.g. interdisciplinary researchers, software engineers, knowledge brokers, and institutional support). Fellowships and career development schemes are essential to strengthen these capacities.
- **Bridging policy to practice**: Translating high-level ethical principles into research workflows requires practical tools, training, and communities of practice to support researchers in addressing the ethical and technical challenges of AI.
- **Anticipating change**: Policy mechanisms that incorporate foresight activities, stakeholder engagement, and continuous learning help keep governance responsive to evolving technologies and practices.

Drawing on insights shared during the MLE process, the infographic below presents a framework for AI in science policy, illustrating the range of policy levers available to support the adoption of AI in research.

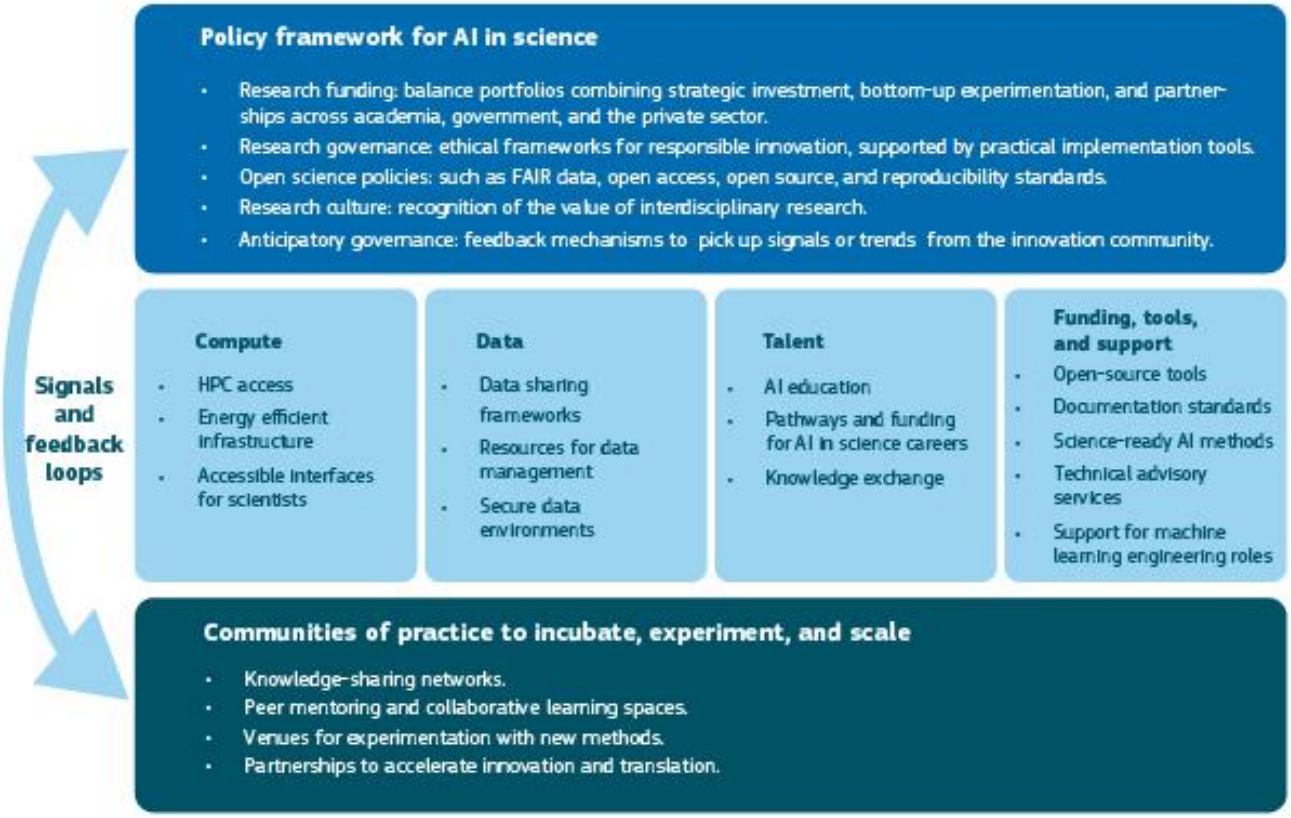


Figure 1: A framework for AI in science policy

1. Introduction

This is the final report of the Mutual Learning Exercise (MLE) on **National Policies for Artificial Intelligence (AI) in Science** that was carried out from July 2024 to June 2025.

AI is changing the landscape of science, offering tools that can accelerate discovery, uncover novel insights in data, and support more efficient, data-driven research. Today, AI applications span all areas of science, technology, engineering, and mathematics (STEM), as well as many disciplines within the arts, social sciences, and humanities. As its capabilities continue to grow, AI presents strategic opportunities, not only to advance scientific leadership, but also to generate economic value and address complex challenges across e.g., health, the climate and security.

The MLE aimed to facilitate the exchange of information, experiences, and lessons learned, and to identify good practices for AI in science policies. It targeted national policy makers and funding agencies, focusing on exchanges of national practices at the operational and strategic levels. It also explored synergies between national strategic approaches across EU Member States and Horizon Europe Associated Countries, while strengthening national capacities to adapt to emerging challenges in AI applications for science.

The MLE was structured around three main topics:

Topic 1: Framework conditions and funding for AI in science

Topic 2: Infrastructures for AI in science

Topic 3: Talent for AI in science

For each topic, participating countries and experts held face-to-face meetings in:

- Brussels, Belgium (Topic 1: 20-21 March 2025)
- Pisa, Italy (Topic 2: 21-22 November 2024)
- Oslo, Norway (Topic 3: 6-7 February 2025)

Participating countries

Austria, Belgium, Estonia, Germany, Hungary, Ireland, Italy, Netherlands, Norway, Romania, and Slovenia.

Contribution to EU policy framework

This report connects with some EU strategic initiatives that increasingly recognise AI as a key priority for Europe's R&I agenda. Thus, it contributes to the content and discussions around ongoing European Research Area (ERA) Actions on AI in science, helping to ensure that national strategies are embedded in a coherent European R&I framework.[1]

A summary factsheet and three thematic reports are available on the dedicated MLE website.

2. Objectives, approaches and structure of the report

This report contains a snapshot of how participating countries are addressing AI in science and is meant to identify opportunities for collaboration, knowledge-sharing, and policy alignment across

[1] The European Research Area (ERA) Actions are specific policy measures and initiatives designed to build a unified research area across the EU https://european-research-area.ec.europa.eu/.

countries. Given the rapid advancement in AI, countries are adopting different strategies to integrate AI into research and their science systems. This report aims to compare policies, identify ongoing practices, and suggest pathways for cooperation.

The report provides a menu of policy recommendations, in some measure calibrated according to each country's stage of AI development and adoption in science. Indeed, as documented in Sections 3 and 4, some countries are already at the forefront of AI adoption in science, while others are still building foundational capacities. Thus, the recommendations provided here are structured to be adaptive rather than one-size-fits-all, so that countries can tailor policies and spot opportunities at EU-level based on their current capabilities and strategic priorities.

The policy menu proposed in this report is not prescriptive; it does not dictate specific actions that must be taken in every situation. Instead, it offers a general framework and guidelines and allows for flexibility in implementation. Nor does it assign priorities to one policy over another, as countries are best positioned to determine priorities based on their specific context, needs, and stage of development (see Section 6).

The report is also relevant for the policy agenda of the ERA and can inform the content and the discussions of all AI-related ERA Actions and especially a future dedicated ERA Action on AI in Science. The aim is to ensure that national AI strategies are contextualised within the larger European AI R&I ecosystems.

2.1. A multi-level approach to policy for AI in science

The MLE discussed national policies for AI in science, yet these are closely interconnected with policies and actions at other levels. In this report, policies and interventions are examined and discussed at two levels:

- **National level**: Refers to national policy makers and actors responsible for the design and implementation of AI strategies, funding programmes, and research infrastructure.
- **Institutional level:** Refers to research organisations, including universities and research centres, and their policies to support AI in science research, integration, and collaboration across disciplines.

The two levels of intervention are interdependent, meaning that policies targeting one level can only be effective if supported and reinforced by actions at the other level. For example, interventions that target individual researchers (e.g., AI training and upskilling programme), can only be successful if there is an institutional environment that provides access to AI tools, computing resources, and funding opportunities. Similarly, interventions at the institutional level (e.g., the establishment of centres of excellence) will be effective only if they are endorsed and supported at the national level through strategic policies, funding allocation, and regulatory clarity.

This report therefore proposes an approach to AI policymaking in science, where policies at different levels reinforce rather than compete with each other. A lack of coordination between these levels may lead to fragmentation, inefficiencies, or missed opportunities.

Some policy interventions may naturally span both national and institutional levels (e.g., investments in energy-efficient HPC facilities). In such cases, policies are assigned and discussed at the level where implementation is likely to have the greatest leverage and impact. However, classifying a policy under one level does not exclude its concurrent implementation or relevance at the other level.

Finally, the discussions during the MLE emphasised the importance of the EU in playing a strategic coordinating role by setting common goals, promoting synergies among Member States, supporting cross-border initiatives, and providing strategic funding programmes.

2.2. Anticipatory innovation governance approach

Throughout the report, we adopt an "anticipatory innovation governance" approach to monitor and anticipate the medium to long-term impacts of AI in science.[2]

The approach recognises that AI is evolving rapidly and many of its future implications for science remain uncertain. Thus, unlike traditional policymaking, which follows linear, long-term cycles, anticipatory governance is adaptive, forward looking and continuously evaluated.

Along with specific policy recommendations, the report will also discuss possible future scenarios, pilot projects, and other policy experiments that may help policy makers remain flexible and responsive to technological changes.

2.3. General structure of the report

The remainder of the report is structured as follows:

- **Background to AI in science.** This section offers an overview of key trends, challenges and opportunities related to the use of AI in science. It is grounded in recent scholarly literature, policy reports, and empirical evidence to contextualise the discussion on the adoption and application of AI across the scientific research landscape.
- **Summary of the MLE meetings, thematic reports, and key challenges.** This section outlines the MLE process, including a recap of the main themes that emerged during the meetings and in the corresponding thematic reports, as well as the policy challenges faced by the participating countries.
- **Policy recommendations across thematic areas and governance levels.** This section presents actionable policy recommendations structured across four thematic areas: (i) *framework conditions for AI in science*, (ii) *funding for AI in science*, (iii) *infrastructures for AI in science*, and (iv) *talent for AI in science*. Recommendations are organised by governance level: national and institutional.
- **Policy recommendations tailored to the national stage of AI maturity in science**. This section adapts recommendations based on the maturity (readiness) of AI in scientific research across countries. The national and institutional policies are structured into a framework that accounts for different stages of AI development and adoption in science. This should allow countries to tailor policies and identify opportunities at the EU level according to their capabilities and strategic priorities.
- **Examples of national and institutional policies for AI in science.** Based on the surveys and thematic reports, Annex 1 discusses how selected countries are approaching AI in science and integrating AI into their scientific ecosystems.

[2] Tõnurist, P. and A. Hanson (2020), "Anticipatory innovation governance: Shaping the future through proactive policy making", *OECD Working Papers on Public Governance*, No. 44, OECD Publishing, Paris, https://doi.org/10.1787/cce14d80-en. Further details on the anticipatory innovation governance approach in the context of AI in science can be found in the first thematic report of this MLE: https://data.europa.eu/doi/10.2777/7211107

3. Background to AI in science

3.1. "AI in science": Definition and policy context

AI in science is the application of AI to science. It covers everything from individual scientists using traditional AI or machine learning (ML) techniques, deep learning, large language models (LLMs) and related foundation models, through to the development of AI technology tailored for scientific needs in various disciplines, and to fully autonomous AI systems - "AI scientists" - that can conduct scientific research on their own. It also includes more basic but widespread use cases of AI-powered tools by scientists as "research assistants" (e.g., literature review, coding, drafting manuscripts, and brainstorming research ideas).

AI is making a significant impact across the sciences, leaving few fields untouched, from materials science and drug design to quantum physics, medicine, and the social sciences and humanities (SSH).[3] Traditionally, AI systems were narrow in scope, and adoption among scientists was limited due to computational constraints and a lack of user-friendly tools. For example, early AI applications in chemistry relied on rule-based expert systems, which had limited predictive power and required extensive manual input. In the early 2010s, breakthroughs in deep learning demonstrated how AI could autonomously generate solutions to complex scientific problems, far exceeding traditional computational methods. Around the same time, AI began expanding into various application domains. More recently, advanced models such as Google's Gemini, OpenAI's GPT-4, and DeepSeek have made AI more accessible, flexible, and capable of handling diverse types of scientific data. In just a few years, AI models have become integrated into scientific workflows, assisting researchers in virtually everything from automating literature reviews to designing experiments and analysing complex datasets.[4]

AI can dramatically shorten research timelines, reducing processes that once took years to just months. It can reduce the manual – and to a lesser extent the intellectual – workload for scientists.[5] Yet as the role of AI in scientific discovery grows, so do concerns about its potential downsides, from risks to scientific integrity and reproducibility to broader questions about how AI might reshape the very nature of science.[6]

Could AI ever replace human intuition and creativity in science? Will it go beyond pattern recognition and incremental findings to propose entirely new paradigms of science? What skills will future scientists need in an AI-driven world? Will AI lead to science without scientists? Although

[3] DeepMind (2024), A new golden age of discovery, https://deepmind.google/public-policy/ai-for-science/; Xu, Y., Liu, X., Cao, X., Huang, C., Liu, E., Qian, S., ... & Zhang, J. (2021). Artificial intelligence: A powerful paradigm for scientific research. *The Innovation*, *2*(4). 10.1016/j.xinn.2021.100179; Hajkowicz, S., Sanderson, C., Karimi, S., Bratanova, A., & Naughtin, C. (2023). Artificial intelligence adoption in the physical sciences, natural sciences, life sciences, social sciences and the arts and humanities: A bibliometric analysis of research publications from 1960-2021. *Technology in Society*, 74, 102260. https://doi.org/10.1016/j.techsoc.2023.102260

[4] Wang, H., Fu, T., Du, Y., Gao, W., Huang, K., Liu, Z., ... & Zitnik, M. (2023). Scientific discovery in the age of artificial intelligence. *Nature*, *620*(7972), 47-60. https://doi.org/10.1038/s41586-023-06221-2

[5] Lu, C., Lu, C., Lange, R. T., Foerster, J., Clune, J., & Ha, D. (2024). The AI scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*. https://arxiv.org/abs/2408.06292; Musslick, S., Bartlett, L. K., Chandramouli, S. H., Dubova, M., Gobet, F., Griffiths, T. L., ... & Holmes, W. R. (2025). Automating the practice of science: Opportunities, challenges, and implications. *Proceedings of the National Academy of Sciences*, *122*(5). https://doi.org/10.1073/pnas.2401238121

[6] Birhane, A., Kasirzadeh, A., Leslie, D., & Wachter, S. (2023). Science in the age of large language models. *Nature Reviews Physics*, *5*(5), 277-280. https://doi.org/10.1038/s42254-023-00581-4; Krenn, M., Pollice, R., Guo, S. Y., Aldeghi, M., Cervera-Lierta, A., Friederich, P., ... & Aspuru-Guzik, A. (2022). On scientific understanding with artificial intelligence. *Nature Reviews Physics*, *4*(12), 761-769. https://doi.org/10.1038/s42254-022-00518-3; Messeri, L., & Crockett, M. J. (2024). Artificial intelligence and illusions of understanding in scientific research. *Nature*, *627*(8002), 49-58. https://doi.org/10.1038/s41586-024-07146-0

predicting the long-term impact of AI in science remains a speculative exercise, a policy framework is urgently needed to deliver the benefits that AI offers for science while ensuring that AI-driven science remains transparent, reliable, and aligned with human oversight and ethical principles.

AI in science is prominently positioned within the European Union's political agenda. In the political guidelines[7] of the European Commission (2024-2029), President Ursula von der Leyen underscores the European ambition to "*focus our efforts on becoming a global leader in AI innovation*". Among the key Commission priorities, the guidelines highlight the development of an Apply AI Strategy in key areas and the establishment of an AI Research Council to pool resources. These priorities are reflected in the mandate[8] of the Commissioner for Startups, Research and Innovation, which explicitly includes the development of a dedicated AI in Science Strategy.

Furthermore, AI in science is identified as a priority area for AI applications in the recent policy-setting initiative of the Commission, AI Continent.[9] It also announces that the AI Research Council will be implemented as a Resource for AI Science in Europe (RAISE). As the current report is being compiled, the European Commission is developing both the Strategy for AI in Science and the structural framework for RAISE. Insights and outcomes from the MLE country visits are expected to inform these strategic initiatives.

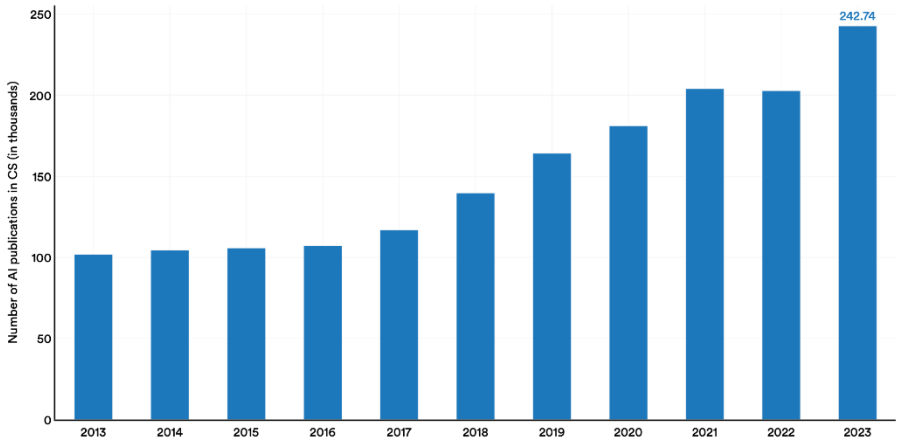
3.2. Empirical evidence on AI in science

3.2.1. Trends in scientific activity

The use of AI for scientific discovery has advanced at pace in the last decades.[10] As documented in the last AI Index Report 2025[11], the number of AI publications continues to grow steadily (Figure 2 – Panel A). Between 2013 and 2023, the total number of publications nearly tripled, rising from approximately 100,000 in 2013 to about 250,000 in 2023. ML has grown the fastest, with publications increasing almost sevenfold since 2015, the year of real take-off. Academia remains the leading global source of AI research, contributing to around 80% of publications over the past decade across all regions (Panel B). However, when it comes to releasing frontier AI/ML models, the balance has shifted: until 2014, academia was in the lead, but since then, industry has taken over. In 2024 alone, industry produced 55 notable ML models. Developing cutting-edge AI models now require massive amounts of data, computing power, and financial resources that are often not available in academia. Still, collaboration between industry and academia remains important.

[7] Political guidelines of the European Commission: e6cd4328-673c-4e7a-8683-f63ffb2cf648_en
[8] Mission Letter to Commissioner Zaharieva: 130e9159-8616-4c29-9f61-04592557cf4c_en
[9] AI continent - European Commission
[10] European Commission: Directorate-General for Research and Innovation, Arranz, D., Bianchini, S., Di Girolamo, V. and Ravet, J., *Trends in the use of AI in science – A bibliometric analysis*, Publications Office of the European Union, 2023, https://data.europa.eu/doi/10.2777/418191; Bianchini, S., Müller, M., & Pelletier, P. (2022). Artificial intelligence in science: An emerging general method of invention. *Research Policy*, *51*(10), 104604. https://doi.org/10.1016/j.respol.2022.104604
[11] Artificial Intelligence Index Report 2025. https://hai.stanford.edu/ai-index/2025-ai-index-report

Panel A. Number of AI publications, 2013–23



Panel B. AI Publications (% of total) by sector and geographic areas, 2023

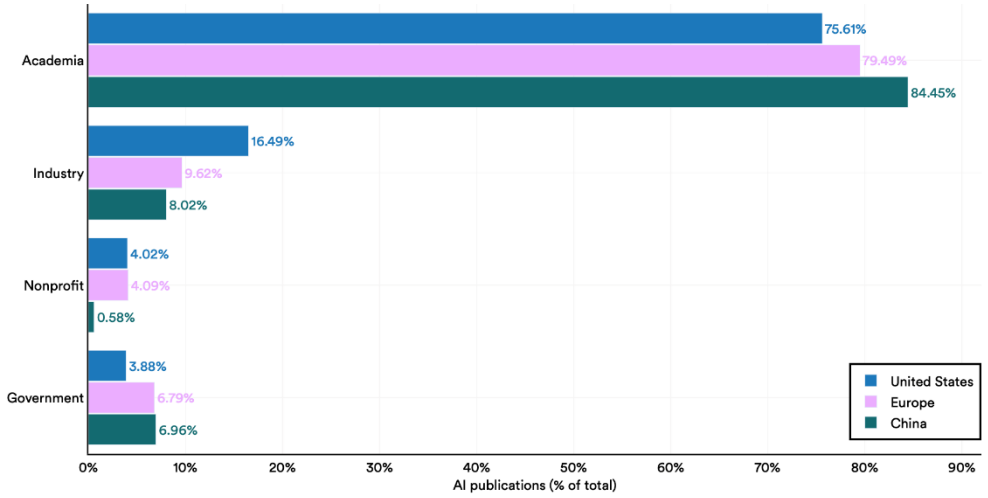
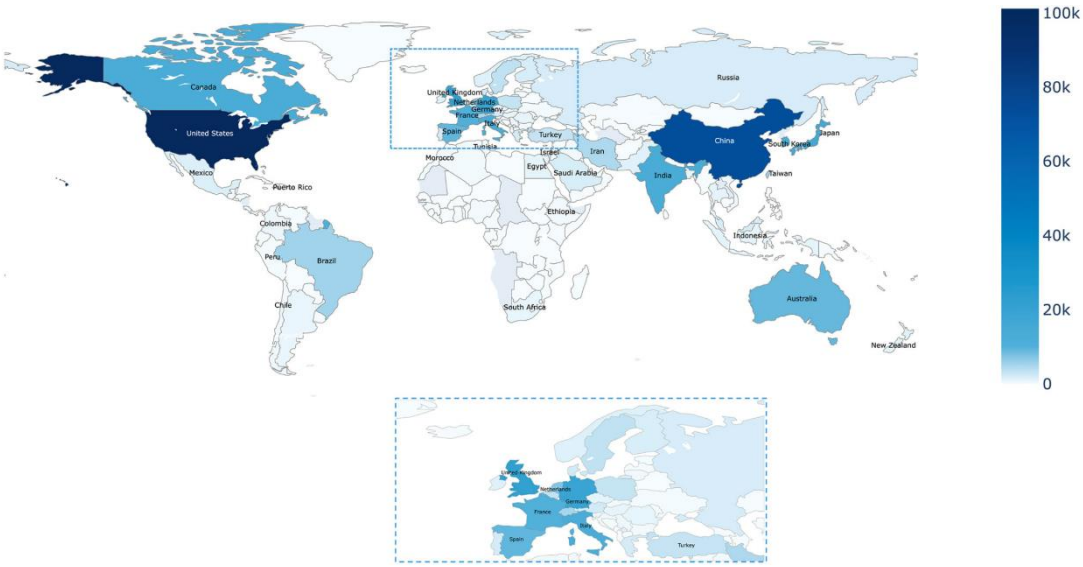


Figure 2: Trends in AI scientific publications
Source: Adapted from AI Index Report (2025)

3.2.2. Global geography of AI-powered research

The race for global leadership in AI is intensifying, including in science. Countries are investing heavily in national AI strategies to guide and foster its development and deployment across scientific disciplines and economic sectors.[12]

Studies on the geography of AI research reveal a very concentrated landscape: China and the United States have come to dominate the AI research system in terms of funding, active scientists, and the number of publications, while Europe follows closely behind. Countries in Africa and Latin America show moderate to low involvement in AI research (Figure 3).[13]



Notes: Cumulative count of AI-focused life science articles by country from 2000 to 2022. Source: Schmallenbach et al. (2024). For figures covering different time periods and scientific fields, see, e.g., Arranz et al. (2024) and AlShebli et al. (2024).

Figure 3:Geography of the AI life science research

It is important to note that rankings are highly sensitive to factors such as scientific disciplines, core vs. applied AI, time period, and – most critically – how AI itself is defined. Most studies confirm that China's research output has surged in recent years, surpassing both the EU and the United States in volume. Meanwhile, the EU and the United States have maintained a steady upward trend. However, the United States and Europe continue to dominate AI research published in high-ranking

[12] European Commission (2021). *Coordinated Plan on Artificial Intelligence 2021 Review*. Brussels: European Commission. https://digital-strategy.ec.europa.eu/en/library/coordinated-plan-artificial-intelligence-2021-review; Wu, F., Lu, C., Zhu, M., Chen, H., Zhu, J., Yu, K., ... & Pan, Y. (2020). Towards a new generation of artificial intelligence in China. *Nature Machine Intelligence*, 2(6), 312-316. https://doi.org/10.1038/s42256-020-0183-4
For an overview of the over 1,000 AI policy initiatives launched by various countries, territories, and the EU, visit the OECD AI Policy Observatory (OECD, 2021). It is worth noting that "AI in science" has generally received only peripheral attention in these plans and policies, as the focus has mainly been on broader economic and social applications of AI technologies.

[13]Klinger, J., Mateos-Garcia, J., & Stathoulopoulos, K. (2021). Deep learning, deep change? Mapping the evolution and geography of a general-purpose technology. *Scientometrics*, *126*, 5589-5621. https://doi.org/10.1007/s11192-021-03936-9

outlets (i.e. top journals) and account for a higher number of citations.[14] A few countries have become hubs of collaboration in AI research. The United States remains central to many of the top international collaborations in Nature Index AI research (2024), as measured by their bilateral collaboration score. There is evidence that other countries are also nodes of activity, including the UK and Germany. China is relatively decoupled from global networks, maintaining a more self-contained research ecosystem. Yet, it seems that China and the United States produce more impactful AI research when collaborating together.[15]

Within Europe, AI research and adoption vary significantly by country. There are leading AI research hubs, for example in the UK, Germany, France, and Italy, while smaller countries and Eastern European countries generally lag behind in research output, although some of these nations are emerging in niche AI applications.[16]

3.2.3. Drivers and barriers to AI adoption in science

Despite AI's growing role in science, several obstacles still prevent many researchers from successfully integrating it into their research. A recent survey of more than 1,600 researchers highlighted that AI adoption is closely tied to understanding its potential within a specific domain, individual skills, and access to computing resources and data.[17] Among scientists who have already experimented with AI, the most commonly cited barriers to wider adoption include: 'Lack of skills or skilled researchers' (80%), 'Lack of training resource' (70%), 'Lack of funding' (50%), 'Lack of computing resources' (35%), and 'Lack of data' (30%). In contrast, scientists who have no prior experience with AI mostly declared that they do not see its relevance to their research.

Empirical research further confirms that the decision to adopt AI depends on both institutional and individual factors. Studies show that adoption is influenced by institutional norms (e.g. proactive AI strategies), the available resources (e.g. HPC facilities), the knowledge and expertise of research collaborators, and personal traits such as pre-existing knowledge and skills, and a taste for experimentation.[18]

Thus, while AI has the potential to transform scientific discovery, its large-scale adoption and effective use depend critically on policies that address skill gaps, training opportunities, infrastructure limitations, and funding constraints (see the conceptual framework in Figure 4).[19] All these aspects were discussed in the MLE process.

[14]See, e.g., the updated statistics on the number of highly-cited AI publications in the 2025 AI Index Report (p.39): https://hai.stanford.edu/ai-index/2025-ai-index-report

[15] Schmallenbach, L., Bärnighausen, T. W., & Lerchenmueller, M. J. (2024). The global geography of artificial intelligence in life science research. *Nature Communications*, *15*(1), 7527. https://doi.org/10.1038/s41467-024-51714-x ; AlShebli, B., Memon, S. A., Evans, J. A., & Rahwan, T. (2024). China and the US produce more impactful AI research when collaborating together. *Scientific Reports*, *14*(1), 28576. https://doi.org/10.1038/s41598-024-79863-5

[16] European Commission: Directorate-General for Research and Innovation, Arranz, D., Bianchini, S., Di Girolamo, V. and Ravet, J., *Trends in the use of AI in science – A bibliometric analysis*, Publications Office of the European Union, 2023, https://data.europa.eu/doi/10.2777/418191

[17] Van Noorden, R., & Perkel, J. M. (2023). AI and science: What 1,600 researchers think. *Nature*, *621*(7980), 672-675. https://doi.org/10.1038/d41586-023-02980-0

[18] Bianchini, S., Müller, M., & Pelletier, P. (2024). Drivers and barriers of AI adoption and use in scientific research. *arXiv preprint arXiv:2312.09843*. https://arxiv.org/abs/2312.09843

[19] For a discussion on the role of compute capacity and corresponding policy actions see: OECD (2023), "A blueprint for building national compute capacity for artificial intelligence", *OECD Digital Economy Papers*, No. 350, OECD Publishing, Paris, https://doi.org/10.1787/876367e3-en.

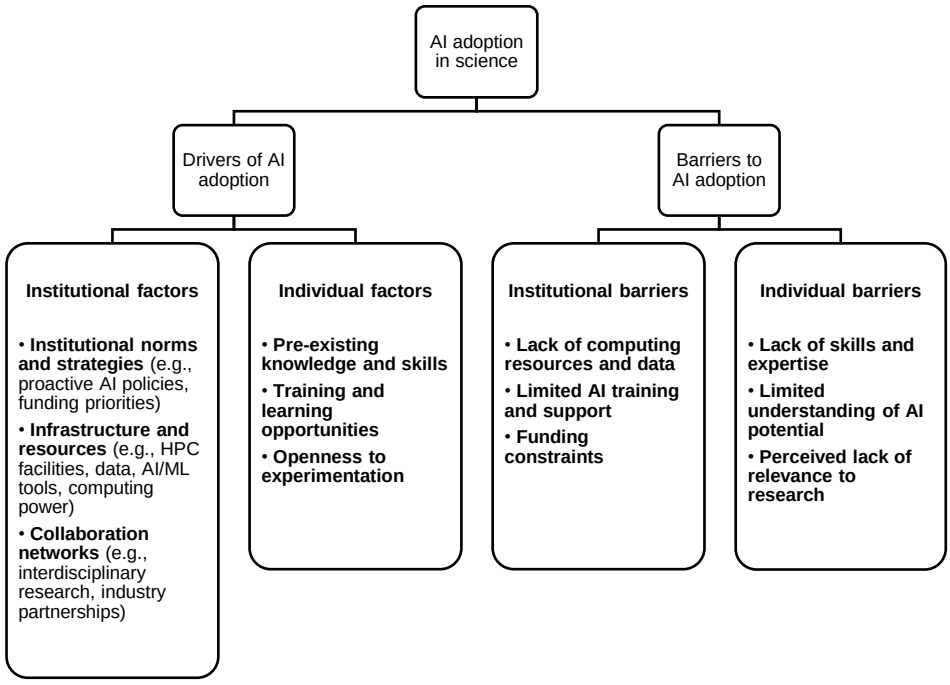


Figure 4: Drivers and barriers to AI adoption in scientific research
Source: Author's elaboration

3.2.4. Mobility of AI talents

AI-driven science requires a unique blend of expertise in fields like computer science, mathematics, statistics and domain-specific knowledge. This makes the cultivation of talent in AI in science particularly challenging. Researchers who can combine technical AI expertise with an understanding of complex scientific problems are rare but essential.

The global competition for AI talent is intensifying[20] and private sector investment in AI in science is growing rapidly. Tech companies invested approx. $189.2 billion (171.8 billion EUR) in AI R&D in 2023.[21] Companies like Google DeepMind, Microsoft Research, and Meta AI are establishing dedicated scientific AI research divisions and pursuing collaborations with academic institutions. With industries offering lucrative opportunities and better infrastructure, many researchers are drawn away from academia and public institutions, leading to a talent drain in the scientific sector.

Studies have shown that young, highly cited scholars from top institutions are the most likely to transition to the private sector and that, after making the switch, their research tends to exhibit lower novelty and impact.[22] Moreover, the industry-driven shift in AI research raises concerns about the marginalisation of exploratory science in favour of commercial interests and the potential misalignment with societal goals.

[20] Ahmed, N., Wahed, M., & Thompson, N. C. (2023). The growing influence of industry in AI research. *Science*, 379(6635), 884-886. https://doi.org/10.1126/science.ade2420
[21] Stanford University (2024), Artificial Intelligence Index Report 2024, Stanford HAI, https://aiindex.stanford.edu/wp-content/uploads/2024/05/HAI_AI-Index-Report-2024.pdf.
[22] Jurowetzki, R., Hain, D. S., Wirtz, K., & Bianchini, S. (2025). The private sector is hoarding AI researchers: What implications for science? *AI & Society*, https://doi.org/10.1007/s00146-024-02171-z

The long-term consequences for science and knowledge production remain uncertain, but proactive policies are needed to ensure competitive career opportunities, funding, and infrastructure for AI researchers within academia and the public sector.

3.2.5. Impact of AI on scientific discovery

AI offers unprecedented capabilities to accelerate discovery – whether by processing vast amounts of data more efficiently to detecting relationships, trends, or anomalies that might elude human researchers (saving time and resources), or by supporting research tasks like synthetising literature, brainstorming ideas, designing experiments, writing code, and more.

Anecdotal evidence from a handful of groundbreaking models and success stories (e.g. DeepMind AlphaFold) exists,[23] yet large-scale empirical evidence on how AI affects scientific productivity and knowledge creation remains limited. Studies generally show that the integration of AI in research results in a citation premium, with AI-related papers appearing among the most highly cited works within and outside their disciplines.[24]

There is substantial heterogeneity in the direct use and potential benefits of AI across different disciplines, but almost every discipline includes some sub-fields that see great benefits from AI.[25] The degree of AI penetration and the structure of knowledge within the field seem to be critical factors in determining its ultimate impact. Fields that are distant from AI tend to benefit more from using AI to advance their research, as do fields where the knowledge space is vast and sparse.[26]

Alongside its epistemic benefits, AI carries also epistemic risks. Over-reliance on AI raises several concerns about its impact on scientific novelty, credibility, research integrity, and fundamental scientific understanding.[27] For instance, the limited empirical evidence on the effect of AI on scientific novelty points to mixed results. We should not exclude a priori the possibility that the widespread adoption and misuse of AI may lead scientists to produce more – but less novel – science and ultimately understanding less.[28]

[23] Yu, Z. (2024). The impacts of prediction AI on scientists: Evidence from AlphaFold. *Available at SSRN 4711334*. https://ssrn.com/abstract=4711334; Toner-Rodgers, A. (2024). Artificial intelligence, scientific discovery, and product innovation. *arXiv preprint arXiv:2412.17866*. https://arxiv.org/abs/2412.17866

[24] Chen, Q., Ho, Y. J. I., Sun, P., & Wang, D. (2024). The philosopher's stone for science–The catalyst change of AI for scientific creativity. (March 5, 2024). https://ssrn.com/abstract=4749118; Duede, E., Dolan, W., Bauer, A., Foster, I., & Lakhani, K. (2024). Oil & water? Diffusion of AI within and across scientific fields. *arXiv preprint arXiv:2405.15828*. https://arxiv.org/abs/2405.15828

[25] Gao, J., & Wang, D. (2024). Quantifying the use and potential benefits of artificial intelligence in scientific research. *Nature Human Behaviour*, *8*(12), 2281-2292. https://doi.org/10.1038/s41562-024-02020-5

[26] European Commission: Directorate-General for Research and Innovation, Bianchini, S., Di Girolamo, V., Ravet, J. and Arranz, D., *Artificial intelligence in science – Promises or perils for creativity?*, Publications Office of the European Union, 2025, https://data.europa.eu/doi/10.2777/6693925

[27] OECD (2023), "AI language models: Technological, socio-economic and policy considerations", *OECD Digital Economy Papers*, No. 352, OECD Publishing, Paris, https://doi.org/10.1787/13d38f92-en

[28] Krenn, M., Pollice, R., Guo, S. Y., Aldeghi, M., Cervera-Lierta, A., Friederich, P., ... & Aspuru-Guzik, A. (2022). On scientific understanding with artificial intelligence. *Nature Reviews Physics*, *4*(12), 761-769. https://doi.org/10.1038/s42254-022-00518-3; Messeri, L., & Crockett, M. J. (2024). Artificial intelligence and illusions of understanding in scientific research. *Nature*, *627*(8002), 49-58. https://doi.org/10.1038/s41586-024-07146-0

4. Topics of the MLE and challenges

This section presents the highlights of the MLE process, offering a recap of the main themes that emerged during the meetings and in the corresponding thematic reports, as well as the policy challenges identified by the participating countries.[29]

The overall landscape reveals a strong heterogeneity in the stage of maturity of AI in science, among participating countries. Gaps span multiple dimensions: infrastructure, funding, governance frameworks, and human capital. Only a few countries currently have a clear and structured vision for AI in science. In most cases, the scientific use of AI is still peripheral to national AI strategies, which tend to prioritise AI for (short-term) economic competitiveness.

EU-wide initiatives are taking shape (see, e.g. EuroHPC JU and the related AI Factories[30] with plans to be even further upgraded in computational power to AI GigaFactories) yet several countries under-exploit these resources and opportunities, often due to limited administrative capacity and low awareness among research institutions. Without actionable national and EU-level policies, disparities in scientific competitiveness are likely to widen. This calls for coordinated EU guidance and benchmarking tools to support national roadmap development – a process for which this MLE has laid important groundwork. These aspects are further elaborated in Sections 5 and 6.[31]

4.1. Framework conditions and funding for AI in science

The MLE meeting on topic 1: framework conditions and funding for AI in science focused on key differences in how countries define and regulate AI in science. Participants first explored enablers and barriers to AI adoption and the institutional interventions needed to address them. There is a collection of research practices and technical developments that can support the use of AI in science, including the availability of AI methods that are fit for purpose, the ability to share best practices through communities of practice, training and capability building activities, career pathways for interdisciplinary researchers, and open software toolkits.

Participants examined more closely the current landscape of funding mechanisms, including direct public funding, competitive grants, and the involvement of the private sector. The discussion also touched on the role of public-private partnerships. The discussion led to the conclusion that funding for AI in science remains highly fragmented and that major disparities exist between countries in terms of investment levels and access to resources. Some countries have dedicated AI research programmes with substantial funding, while others rely heavily on private sector investment, leading to concerns about long-term sustainability and equitable access to AI infrastructure. Some countries leverage industry collaborations to enhance AI research and infrastructure, while others expressed concerns over dependency on private investments, particularly regarding data access, ownership, and the influence of commercial interests on research priorities. The discussion also highlighted tensions between bottom-up and top-down approaches to funding AI in science, and

[29] A summary factsheet and three thematic reports are available on the dedicated MLE webpage.

[30] The European High-Performance Computing Joint Undertaking (EuroHPC JU) is a public-private partnership established in 2018 to develop a world-class supercomputing ecosystem in Europe. It pools resources from the EU, participating countries, and private partners to enhance Europe's capabilities in HPC, quantum computing, and AI. AI Factories leverage the supercomputing capacity of the EuroHPC JU to develop trustworthy cutting-edge generative AI models. For more details: AI Factories | Shaping Europe's digital future

[31] See also: European Commission: Directorate-General for Research and Innovation, Petkova, D. and Roman, L., *AI in science – Harnessing the power of AI to accelerate discovery and foster innovation – Policy brief*, Publications Office of the European Union, 2023, https://data.europa.eu/doi/10.2777/401605

the need to strike a balance between investment in technical infrastructure and support for human capital.

Finally, the meeting focused on frameworks for the responsible use of AI in science, touching upon the EU's Principles for Trustworthy AI,[32] the evolving Guidelines on the Responsible Use of Generative AI in Research,[33] the EU Artificial Intelligence Act,[34] and the upcoming Council Conclusions on AI in Science, initiated by the Polish Presidency of the Council of the EU. Discussions highlighted that while these frameworks provide an important foundation, there is still uncertainty about their practical implementation in scientific research and in various national contexts. A recurring theme was the need for clearer, more harmonised guidelines for European research institutions to ensure compliance while maintaining flexibility for scientific exploration.

The need for dialogue between AI policy and scientific practice emerged as a key issue, particularly in governance, ethics, and funding. Participants highlighted the importance of practical interventions that help researchers implement existing governance frameworks within their workflows.

The key challenges that emerged from discussions are reported in Figure 5. The thematic report is available here: European Commission: Directorate-General for Research and Innovation and Montgomery, J., *Framework conditions and funding for AI in science* – Mutual learning exercise on national policies for AI in science – First thematic report, Montgomery, J.(editor), Publications Office of the European Union, 2025, https://data.europa.eu/doi/10.2777/7211107

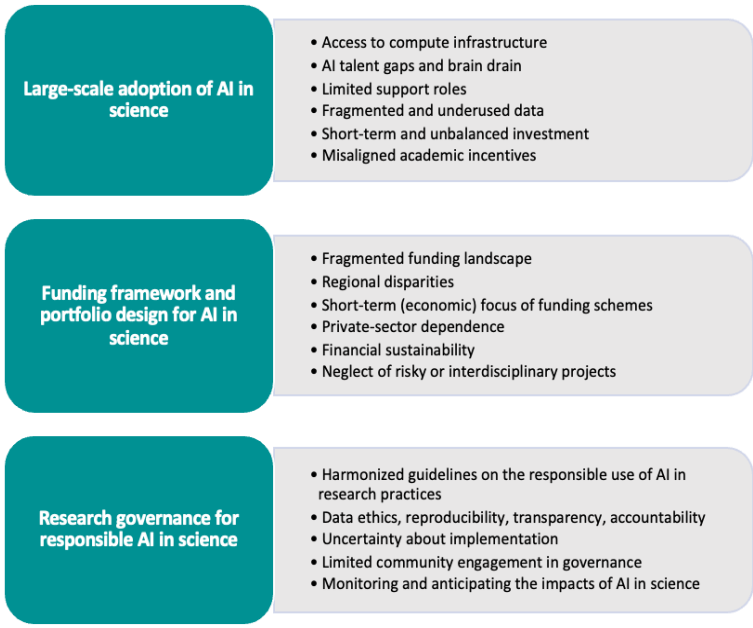


Figure 5: Key challenges identified in the MLE process for framework conditions and funding for AI in science
Source: Author's elaboration

[32] European Commission (2019). *Ethics Guidelines for Trustworthy AI*. High-Level Expert Group on Artificial Intelligence. Brussels: European Commission. Available at: https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai

[33] European Commission (2024). *Guidelines on the Responsible Use of Generative AI in Research*, developed by the ERA Forum. Brussels: European Commission. Available at: https://research-and-innovation.ec.europa.eu/news/all-research-and-innovation-news/guidelines-responsible-use-generative-ai-research-developed-european-research-area-forum-2024-03-20_en

[34] https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai

4.2. Infrastructures for AI in science

The MLE meeting on topic 2: infrastructures for AI in science focused on the availability, accessibility, and sustainability of AI-related research infrastructures across different countries. These infrastructures include computing resources, data access, storage and connectivity, security, scalability, ethical compliance, and cost management.

Participants explored the challenges of ensuring that researchers have access to the necessary computing power, data resources, and software tools to fully leverage the potential of AI in scientific research. A key conclusion was the unequal distribution of AI infrastructure: while some countries and institutions benefit from advanced HPC facilities, others struggle with limited access to computational resources and cloud-based AI tools. Cost barriers, technical capacity and educational gaps, and regulatory constraints were identified as the main obstacles to equitable access. Moreover, commercial firms dominate computing resources for AI in science.

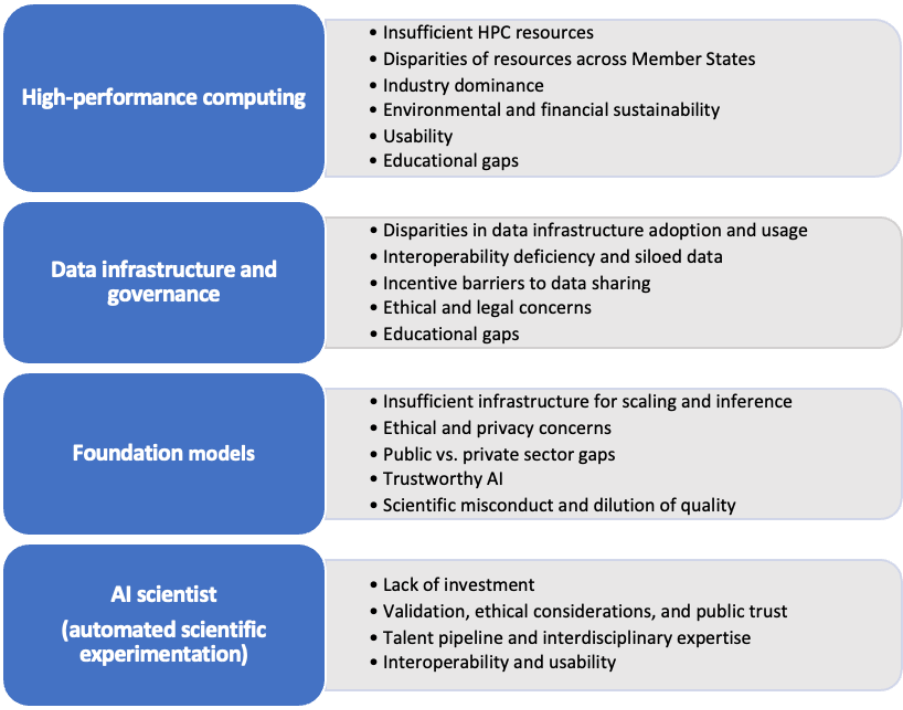


Figure 6:Key challenges identified in the MLE process related to infrastructures for AI in science
Source: Author's elaboration

A major issue that emerged from the meeting was the standardisation and interoperability of AI infrastructures across different research institutions and countries. The lack of common technical standards, metadata frameworks, and data-sharing protocols creates inefficiencies and limits the scalability of AI applications in science. Participants emphasised the importance of developing European-wide strategies to ensure that AI infrastructures are interoperable, secure, and accessible to a broad range of researchers.

Another theme of discussion was the role of foundation models in science. Participants highlighted challenges such as high computational costs, dependence on proprietary models developed by private companies, and concerns about reproducibility and bias. The debate also addressed whether publicly funded AI infrastructures should prioritise the development of open-source foundation models to ensure broader accessibility and transparency in AI-driven science.

Participants agreed on the importance of adhering to FAIR principles (Findable, Accessible, Interoperable, and Reusable) for scientific data management.[35]

Finally, the meeting explored the emerging role of AI scientists – i.e. fully autonomous AI-driven laboratory systems capable of designing, executing, and analysing experiments without direct human intervention. While these systems promise to accelerate discovery, concerns were raised about how to integrate them effectively into existing research environments, the potential impact on scientific jobs and training, and the ethical implications of delegating experimental decision-making to AI.

The key challenges that emerged from discussions are reported in Figure 6.[36] The thematic report is available here: European Commission: Directorate-General for Research and Innovation and King, R., *Infrastructures for AI in science* – Mutual learning exercise on national policies for AI in science – Second thematic report, King, R. (editor), Publications Office of the European Union, 2025, https://data.europa.eu/doi/10.2777/8581577.

4.3. Talent for AI in science

The MLE meeting on topic 3: talent for AI in science focused on the availability, mobility, and training of AI-skilled researchers across different countries. Participants explored the challenges of developing, attracting, and retaining AI talent in academia, public research institutions, and industry.

A key conclusion was the growing competition for AI expertise. Many skilled researchers are increasingly drawn to the private sector due to higher salaries, better infrastructure, and greater access to cutting-edge AI technologies. This has led to concerns about a brain drain from academia to industry on the one hand, and from Europe to other countries on the other, which could impact the long-term development of AI research and science in Europe. In response, the discussion highlighted the need for a coordinated investment agenda to retain and attract AI talent in academia. This includes the role of funding agencies and institutions in providing long-term career paths, stable funding opportunities, and access to high-quality research infrastructures.

Another major theme was the mobility of AI researchers, both within and across national borders. While many of the issues discussed apply to scientists more broadly, several barriers to mobility were identified in the context of AI, including restrictive visa policies, bureaucratic barriers, and uneven funding opportunities which often limit the movement of talent.

AI education and training were other important topics. Participants highlighted the need for interdisciplinary curricula that integrate computer science, mathematics, statistics, and domain-specific expertise. It turned out that some countries have invested in dedicated AI training programmes at the university and postgraduate levels, while others still face gaps in AI education, particularly in the availability of specialised courses, faculty expertise, and hands-on training opportunities. Participants also highlighted the importance of upskilling researchers who do not have a traditional AI background but increasingly rely on AI tools in their scientific work. Also, there was discussion about introducing AI concepts earlier in the education system, including in primary and secondary education, to prepare future generations for AI-driven science.

[35] Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., Appleton, G., Axton, M., Baak, A., ... & Mons, B. (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 3(1), 1-9. https://doi.org/10.1038/sdata.2016.18

[36] Other infrastructure-related challenges, not necessarily limited to AI, are discussed in the report Towards a European Policy for Technology Infrastructures – Building Bridges to Competitiveness (European Commission, Directorate-General for Research and Innovation, Publications Office of the European Union, 2025): https://data.europa.eu/doi/10.2777/0876395

The discussion also emphasised interdisciplinarity as a key factor. AI research and research with AI increasingly require collaboration between computer scientists, domain experts, lawyers, and ethicists, yet many academic structures and funding models remain discipline specific. Participants noted that AI scientists in application domains often struggle to find career opportunities that bridge multiple fields, as hiring, tenure, and promotion systems still tend to favour traditional disciplinary boundaries.

Participants also considered the role of diversity and inclusion in AI research and AI in science, as the field continues to face gender imbalances and underrepresentation of minority groups. Strategies to promote diversity in AI careers were discussed, including targeted scholarships, mentorship programmes, and institutional policies that encourage more equitable hiring practices.

Finally, discussions touched on the ethics and societal responsibility of AI researchers. Participants agreed that AI training programmes must also address ethical concerns, such as bias, fairness, and the societal and legal implications of AI deployment in science.

The key challenges that emerged from discussions are reported in Figure 7. The thematic report is available here: European Commission: Directorate-General for Research and Innovation and Groznik, V., *Talent for AI in science* – Mutual learning exercise on national policies on AI in science – Third thematic report, Groznik, V. (editor), Publications Office of the European Union, 2025, https://data.europa.eu/doi/10.2777/8779512.

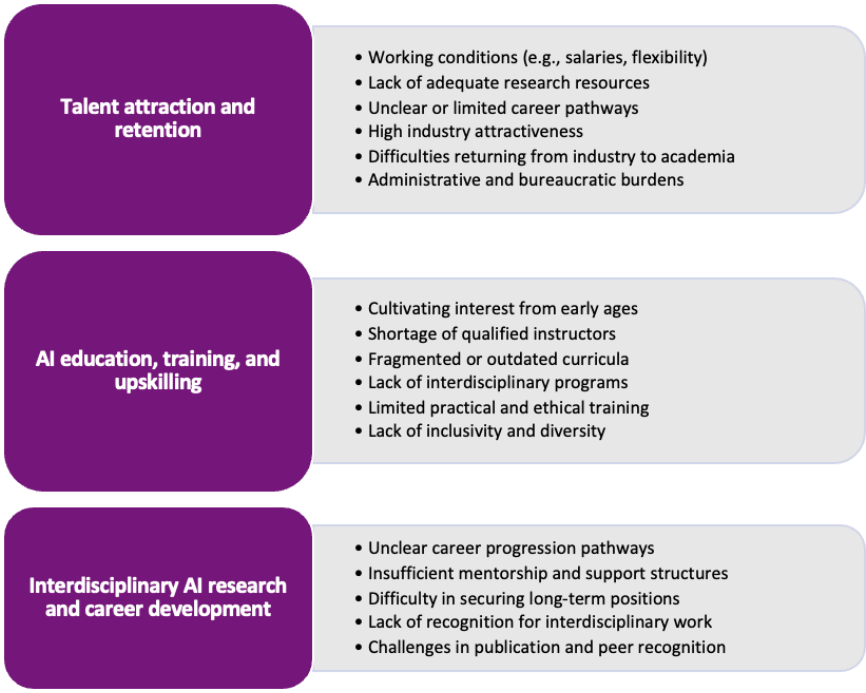


Figure 7:Key challenges identified in the MLE process related to talent for AI in science
Source: Author's elaboration

5. Policy recommendations for AI in science by thematic area and governance levels

This section presents a menu of actionable policy recommendations, for policy makers at national level and for research organisations. The recommendations are structured by the four thematic MLE areas, namely: (i) **framework conditions for AI in science**, (ii) **funding for AI in science**, (iii) **infrastructures for AI in science**, and (iv) **talent for AI in science**.

5.1. Policy recommendations on framework conditions for AI in science

For national policy makers:

- Develop (and regularly update) national roadmaps and visions for AI in science, which are aligned with European strategies and adapted to local priorities and capabilities.
- Embed AI as a cross-cutting research priority in national R&I policies, supported by targeted funding schemes, fellowships and training programmes.
- Create practical tools (e.g. self-assessment guides, training modules, documentation standards) to operationalise ethical and responsible frameworks for AI in science.
- Coordinate AI policies across departments (AI, research, infrastructure, education) to ensure coherence and efficiency.
- Set up and connect to networks of excellence to share best practices, pool expertise, and align national efforts with European developments.
- Support communities of practice to foster interdisciplinary knowledge-sharing.
- Fund dedicated support structures (e.g. AI advisors, technical consultants) to assist researchers.
- Monitor and evaluate the impact of AI in science with built-in feedback loops.

For research organisations:

- Align institutional strategies with national/EU priorities, while preserving academic freedom and research autonomy.
- Create protected spaces and dedicated fellowship programmes to enable researchers to experiment with AI tools and methods.
- Promote communities of practice within and across institutions.
- Implement FAIR data practices and reward transparency, openness, and good data stewardship.
- Participate in and help build networks of excellence that connect institutions nationally and internationally.

EU role: MLE participants highlighted the importance of the EU in setting the global vision on AI and AI in science and keeping policy coherence across research, infrastructure, and digital strategies.

5.2. Policy recommendations on funding for AI in science

For national policy makers:

- Design balanced funding portfolios that combine top-down investments with bottom-up research-led experimentation, embedding both in a coherent strategic investment agenda.
- Allocate resources to both technical (HPC, software) and human (AI specialists, data managers) infrastructure.
- Support interdisciplinary collaboration with tailored funding, adapted evaluation criteria, fellowships, and joint positions bridging AI and domain expertise.
- Establish long-term, stable funding instruments for risky and interdisciplinary projects on AI in science.
- Promote co-financing models with industry, while safeguarding openness, reproducibility, and the public interest.

For research organisations:

- Offer seed funding and fellowships to support early AI experimentation, especially for early-career researchers.
- Develop internal co-financing mechanisms (e.g., matched funding with industry) that ensure sustainability and autonomy.
- Ensure inclusive access to AI funding across departments and disciplines.

EU role: MLE participants discussed the important role of EU funding programmes for investing in exploratory research and infrastructure for AI in science, and for addressing regional disparities in access to AI capabilities.

5.3. Policy recommendations on infrastructures for AI in science

For national policy makers:

- Increase support for national HPC and data infrastructures and promote participation in EU-level initiatives (e.g. EuroHPC, AI Factories, RAISE), as part of a coordinated investment agenda.
- Incentivise private sector contributions (e.g. data, computing power) to boost AI resource availability.
- Invest in energy-efficient, smaller-scale AI systems and sustainable HPC infrastructure.
- Create clear regulatory frameworks for data governance (e.g. ownership, access rights, GDPR compliance) and responsible AI use in science.
- Invest in national repositories and cloud infrastructures for secure, reliable, and long-term data accessibility.

- Support the development of targeted infrastructure that facilitates scientific discovery and scientific workflow, such as domain-specific foundation models and AI systems for automated scientific work (i.e. robot scientists).
- Promote engagement in European networks of excellence to strengthen cross-border collaboration, harmonise standards, and optimise use of shared infrastructures.

For research organisations:

- Ensure AI-ready infrastructure is accessible, interoperable, and efficiently used.
- Train researchers to use and validate AI models for scientific research; raise awareness on risks of over-reliance.
- Promote cross-disciplinary collaboration on HPC and AI applications, and connect to networks of excellence.
- Standardise data formats and reward contributions to FAIR/open data practices.
- Provide support roles (e.g. AI/data stewards) and training in data management.
- Monitor AI integrity through internal governance and compliance systems.

EU role: Participants of the MLE highlighted the important role of EU investment in scientific infrastructures, specifically data infrastructures, HPC (especially the AI Factories with a focus on science), and funding for foundation models and large-scale initiatives in automated science. They stressed that computational resources must be strategically designed to support both AI training (learning) and deployment (inference), addressing the specific computational needs of AI-driven science and ensuring effective distribution across countries, disciplines, and research communities. The participants recognised the need to expand the European Open Science Cloud (EOSC) to ensure seamless and equitable access to data for researchers. Participants called on the EU to engage in international efforts to anticipate and mitigate risks related to the use of AI in scientific discovery.

5.4. Policy recommendations on talent for AI in science

For national policy makers:

- Include AI fundamentals and computational thinking in primary and secondary school curricula to build early digital literacy.
- Fund AI outreach initiatives in schools and communities to raise awareness from an early age and foster scientific curiosity.
- Support grants, fellowships, and subsidies for interdisciplinary AI training (“AI +discipline X”) and lifelong learning.
- Support national programmes for improving HPC-related skills among researchers.
- Create national AI research centres or integration hubs promoting interdisciplinary collaboration and acting as nodes for international networks of excellence.
- Establish co-financing models where industry partners or EU programmes contribute to research salaries in collaborative AI in science projects.

For research organisations:

- Establish structured career pathways and incentives for researchers applying AI in science (e.g. tenure-track positions with clear progression, performance-based salary schemes, and targeted fellowships).
- Reform evaluation criteria to recognise AI training, interdisciplinarity, and upskilling efforts.
- Enable "dual track" career models to support mobility between academia and industry.
- Expand AI education across scientific disciplines, including mandatory onboarding modules for PhDs and early-career researchers.
- Support interdisciplinary PhD/postdoctoral programmes combining AI with specific research domains (e.g. AI + biology, AI + social science), ideally linked to national or European networks of excellence.
- Deliver discipline-specific training and awareness programmes on AI ethics, reproducibility, and responsible research conduct, providing spaces for critical reflection about the impact of AI in science.
- Embed AI and HPC training into academic curricula to create a skilled, interdisciplinary workforce for hands-on AI training and practical exposure to real-world AI applications.

EU role: MLE participants highlighted the importance of the EU in promoting international mobility and attracting AI talent by simplifying and harmonising visa policies across Member States and Associated Countries. They also stressed the need for Horizon Europe and other EU programmes to support AI-focused career development (e.g., EU-wide postdoctoral fellowships for AI-in-science projects).

6. Policy recommendations tailored to the stage of AI maturity in science

In this section, three stages of "maturity" (or "readiness") of AI in science are proposed (see Figure 8). This taxonomy should be considered as a guiding framework for policy makers and science administrators to assess the AI maturity level of their country or institution. Further work is needed to develop metrics and indicators that provide a more precise evaluation; this task was beyond the scope of this MLE. To streamline the analysis, the topics discussed in the previous sections have been grouped into two macro thematic areas, namely: (i) infrastructures for AI in science and (ii) talent for AI in science. Other topics naturally cut across these four areas (e.g. research governance, AI ethics, funding mechanisms, interdisciplinarity, public-private collaboration) and are therefore addressed within each theme where most relevant.

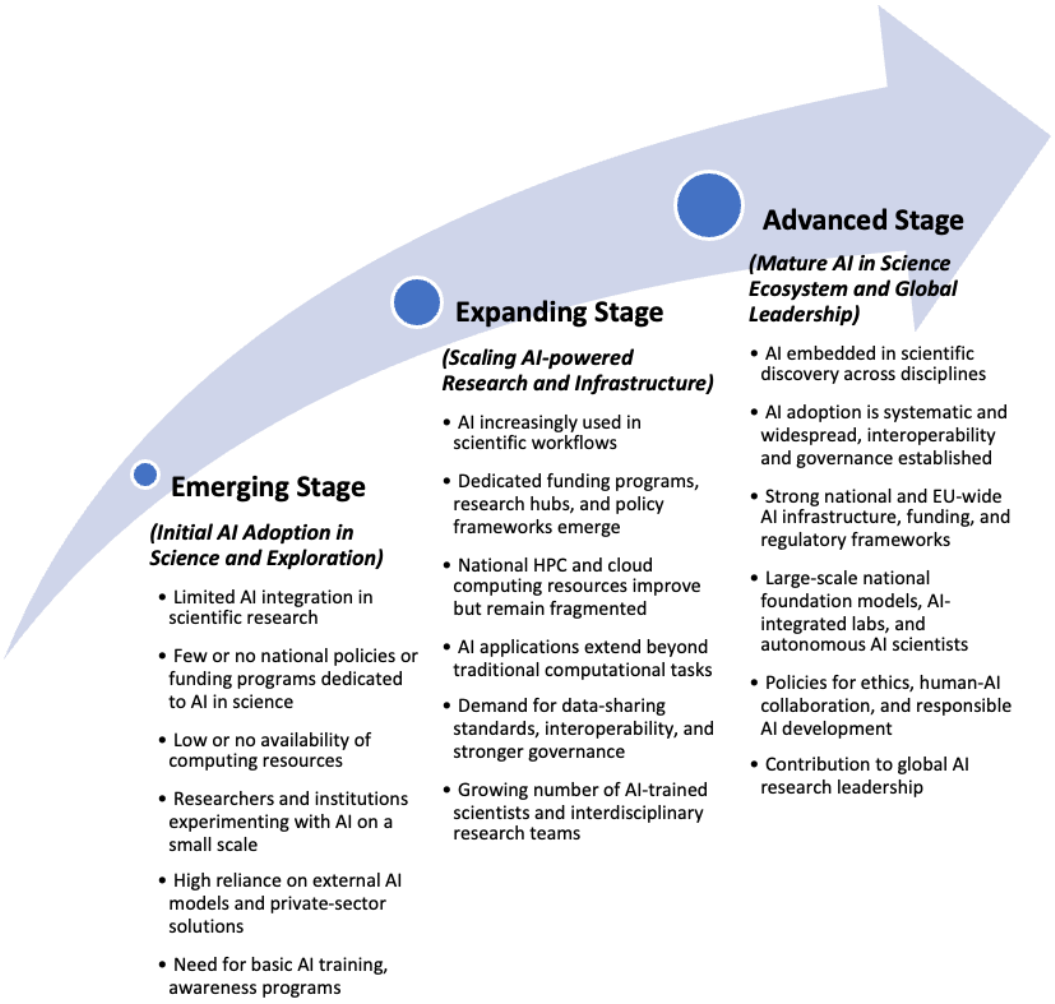


Figure 8: Country readiness for AI in science
Source: Author's elaboration

6.1. Policy recommendations on infrastructures for AI in science

Emerging stage (*initial AI adoption in science and exploration*)

- Develop a national AI strategy (if not yet in place) with a long-term vision for AI in science, explicitly including HPC and data needs - where such a strategic orientation is desired.
- Identify priority research areas that would benefit most from HPC investments and improved data access.
- Promote access to and effective use of EU-level initiatives (e.g. EuroHPC, EOSC, AI Factories) to benefit from shared transnational computing and data resources, and connect with the European network of excellence in AI in science.
- Support international collaboration to gain expertise from countries with more mature HPC and data infrastructures.
- Attract private-sector partnerships to provide cloud-based AI resources where national HPC infrastructure is still lacking or underdeveloped.

- Raise awareness and build foundational skills in HPC and data management through introductory training programmes for researchers and engineers.
- Promote institutional engagement in FAIR data principles and the standardisation of data formats.
- Launch national and/or institutional exploratory programmes to support the initial use of foundation models and AI automation in research.

Expanding stage (*scaling AI-powered research and infrastructures*)

- Increase national funding as part of a strategic investment agenda for dedicated HPC and data infrastructure, ensuring capacity for both AI training and inference.
- Integrate long-term HPC and data planning into national AI strategies to ensure sustainability and coordinated investment.
- Enhance participation in European initiatives by integrating national research infrastructures into EU-wide networks and networks of excellence.
- Encourage private sector engagement by incentivising industry contributions to national computing and data infrastructure (e.g. computing resources, cloud platforms, funding).
- Expand AI workforce training programmes, developing specialised programmes on HPC and data stewardship; offer specialised fellowships to support skill development.
- Establish national incentives for open data sharing, including recognition in funding, promotion, and evaluation criteria.
- Establish AI-integrated labs combining foundation models for science and automation of scientific discovery (i.e. “AI scientists”).

Advanced stage (*mature AI in science ecosystem and global leadership*)

- Expand large-scale, high-performance AI computing clusters as part of a long-term investment agenda to support cutting-edge core AI research and applications of AI in science.
- Establish HPC testbeds to accelerate AI model training, experimentation, and deployment.
- Invest in next-generation HPC technologies (e.g. quantum computing and energy-efficient AI hardware and storage systems) to future-proof infrastructure.
- Develop a sustainability framework to reduce the environmental footprint of HPC and large-scale data use.
- Ensure full interoperability between national and EU-wide HPC systems for seamless cross-border collaboration.
- Foster public-private AI research ecosystems, where private and public sectors co-develop and share computing resources.
- Institutionalise reward systems for open data contributions, reproducible science, and responsible data governance.
- Establish institutional governance bodies to ensure compliance with FAIR principles and conformity with national/EU data strategies.

- Lead international coordination on standards, safety, and openness of foundation models and scientific automation, leveraging European networks of excellence to position Europe at the forefront of global AI in science governance.

6.2. Policy recommendations on talent for AI in science

Emerging stage (*initial AI adoption in science and exploration*)

- Introduce AI basics and computational thinking in primary and secondary school curricula to promote early digital literacy.
- Support AI outreach programmes in schools and communities to promote awareness and foster curiosity about AI from an early age.
- Expand AI education across disciplines in universities and research institutions, beyond computer science.
- Offer onboarding AI modules and entry-level fellowships for PhD students and early-career researchers in all scientific fields.
- Support public engagement and outreach by scientists to demystify AI and encourage informed societal dialogue.
- Support national AI infrastructures and emerging centres of excellence to build strong research environments and prepare for future participation in networks of excellence.
- Simplify hiring and grant access procedures by streamlining administrative processes (i.e. reducing paperwork, shortening approval timelines, and creating clear, standardised procedures for researchers and institutions).
- Raise awareness of AI career opportunities (including in domain sciences) through public engagement campaigns and early-career mobility programmes focused on research roles.

Expanding stage (*scaling AI-powered research and infrastructures*)

- Introduce AI basics and computational thinking in primary and secondary school curricula to promote early digital literacy.
- Provide incentives for universities to create interdisciplinary AI degrees, minors, or micro-credentials.
- Enable flexibility in academic institutions through the development of modular, stackable learning formats and hybrid delivery models.
- Provide release time or teaching incentives for faculties involved in developing and delivering new AI-related content.
- Reform academic evaluation criteria to recognise contributions to AI education, upskilling, and interdisciplinary teaching approaches.
- Strengthen public-private partnerships to support hands-on training, internships, and exposure to applied AI settings.
- Support public engagement and outreach by scientists to demystify AI and AI-enabled science/discovery and encourage informed societal dialogue.
- Introduce mobility-based salary adjustments to offset cost-of-living gaps and support geographical mobility (not limited to AI researchers).

- Establish co-financing models with industry and EU funding programmes to support salaries and joint research appointments in AI collaborative projects.
- Develop structured tenure-track pathways for AI researchers, with criteria valuing interdisciplinary work.
- Support “dual track” academic-industry careers, allowing scientists to contribute to both sectors with contractual protections and clear career prospects.
- Connect institutional efforts to national and European networks of excellence, ensuring knowledge exchange and visibility.

Advanced stage (*mature AI in science ecosystem and global leadership*)

- Institutionalise national frameworks for AI literacy across all education levels, anchored in a long-term investment agenda for digital and scientific capacity.
- Ensure agility and responsiveness in AI education by mandating regular review cycles of curricula and training content.
- Support dedicated national centres of excellence for interdisciplinary AI education and advanced research training, embedded within European and global networks of excellence.
- Lead in setting global standards for inclusive, ethical, and interdisciplinary AI education.
- Support public engagement and outreach by scientists to build trust, critical thinking, and societal readiness for AI.
- Embed AI-specific career incentives into academic reward systems, including promotion, salary progression, research sabbaticals, and leadership roles.
- Reform research evaluation frameworks to explicitly recognise work in AI in science, including training, interdisciplinary collaboration, open data/software contributions, and public communication.

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Annex 1: Overview of policies and programmes for AI in Science in selected Member States and Associated Countries

Member State/ Associated Country	Relevant examples of programmes for AI in Science
Austria	*Policy framework*: • Austria has a **national AI Strategy** "Artificial Intelligence Mission Austria 2030" since 2021. In 2024, the "Implementation Plan 2024" was published, which concretises and supplements the existing AI strategy. Education, science and research are addressed in the strategy. • Under the name **AI Mission Austria** (AIM AT) the three funding agencies AWS, FFG and FWF funded projects in basic and applied research with a total of 12 million EUR for the period 2022-2023. • **FWF Cluster of Excellence Program**: The Bilateral AI research cluster is funded with 19,8 million EUR for a period of 5 years. It combines two of the most important strands of research in the field of AI: sub-symbolic AI (ML) and symbolic AI (knowledge representation and reasoning). • In 2021, the Austrian Federal Ministry for Climate Action, Environment, Energy, Mobility, Innovation and Technology (since 2025 the Federal Ministry for Innovation, Mobility and Infrastructure) launched an **AI for Green programme** to develop AI applications in climate protection and adaptation. • The Austrian Research Promotion Agency (FFG) **"Digital Technologies" programme** addresses among other elements interdisciplinary research projects that leverage AI in research and other sectors. • Within the performance agreements with all Austrian public universities for the period 2025-2027, the Ministry of Women, Science and Research (formerly the Ministry of Education, Science and Research) and the universities agreed on a **number of actions to increase AI use in academia** including: the increase of AI up- and reskilling measures for students, higher education teachers and scientists, new or upgraded institutional digitalisation/AI strategies, the upgrade of HPC-infrastructure, and cybersecurity measures.

Member State/ Associated Country	Relevant examples of programmes for AI in Science
	Talent: • **FWF Stand-Alone Projects** offers funding for interdisciplinary research, including AI adoption in scientific disciplines. • **FWF supports training and development programmes** aimed at reskilling researchers in AI. This includes initiatives where researchers in traditional fields, such as social sciences or natural sciences, are trained in AI techniques to enhance their research capacities. • **AITHYRA** – Newly founded ÖAW Research Institute for Biomedical Artificial Intelligence (co-funded by industry: Boehringer Ingelheim Foundation) aims to develop AI-first approaches to transform biological sciences and to attract AI talent (computer scientists, engineers, and biologists) from abroad.
Belgium (Flanders)	*Policy framework*: • **Open Flemish R&I funding instruments** allow submissions of interdisciplinary AI-research in science proposals. *Infrastructures*: • KU Leuven is advancing AI integration in science through **Leuven.AI**, a centre in Flanders with 500 researchers. A key initiative is **AI as a Service** (AIaaS), which provides AI-research software engineers to support research teams. *Talent*: • **VAIA** accelerates the adoption of AI in Flemish companies by consolidating and distributing a comprehensive range of **AI training courses** for professionals and researchers. • Flanders (Belgium) offers a range of programmes, including Flanders AI Research, Leuven.AI, Delta at University Ghent, the Antwerp Centre for Responsible Artificial Intelligence, and AI@UHasselt. • Flanders AI Research is a dedicated programme funded by the Flemish government (14 million EUR/year).

Member State/ Associated Country	Relevant examples of programmes for AI in Science
Estonia	*Talent*: • Estonia supports interdisciplinary AI-research through the **Estonian Centre of Excellence** in Artificial Intelligence.
France	*Policy framework*: • The French government has committed 54 billion EUR to support companies, universities, and research institutions in strategic technology areas, with AI as a key focus. A cornerstone of this strategy is the **AI-Cluster programme**, which invests 2.22 billion EUR over five years to develop centres of excellence in AI research and education. *Talent*: • France's **National AI Strategy** provides dedicated funding to support scientists in upskilling, ensuring they can effectively integrate AI into their work.
Germany	*Policy framework*: • Germany utilises transdisciplinary research areas, the **Cluster of Excellence 'Science of Intelligence'**, and DFG Research Units in AI to support interdisciplinary collaboration. • Germany's Cyber Valley Initiative combines expertise in AI and robotics with applications in materials science and healthcare, facilitating cross-disciplinary innovation. *Infrastructures*: • Germany's Helmholtz Association has committed 23 million EUR to the development of foundation models for science. The **Helmholtz Munich Foundation Model Initiative (HFMI)** will support four pilot projects aiming to: improve climate modelling through AI; enhance medical imaging for tumour detection; advance photovoltaic material research; and improve carbon cycle modelling.

Member State/ Associated Country	Relevant examples of programmes for AI in Science
	• The University of Tübingen's Cluster of Excellence in Machine Learning for Science brings together university expertise in AI and various scientific disciplines in an environment that encourages interdisciplinary collaboration. *Talent*: • Deutsche Forschungsgemeinschaft's (DFG) strategic funding initiative on AI attracts and retains early-career AI-experts, notably through independent research groups in the **Emmy Noether Programme**, providing early-career AI researchers with flexible funding and independent research positions. • Germany's Helmholtz Information and Data Science Academy (HIDA) ensures that PhD-level interdisciplinary AI training is embedded into scientific disciplines, including life sciences and engineering. HIDA focuses on training PhD students and postdocs in both data science and domain-specific AI applications.
Hungary	*Policy framework*: • HUN-REN is a research network that brings together 11 research centres, 8 research institutes, and 116 supported research groups across Hungary. Its AI for Science initiative aims to promote the use of AI across this network of research institutions. *Talent*: • The HUN-REN AI4Science programme empowers researchers to act as ambassadors, helping promote the use of AI across different scientific disciplines within the HUN-REN research institute network.
Ireland	*Policy framework*: • The Irish Government convenes a collection of **funding programmes to support collaborations between academia and industry**, including Innovation Vouchers, Innovation Partnerships, Technology Centres, and European Digital Innovation Hubs.

Member State/ Associated Country	Relevant examples of programmes for AI in Science
	• **Technology Ireland ICT Skillnet** is a government-backed business support network that connects teams and individuals from both small and large companies to highly subsidised and cutting-edge training in the latest technologies driving business development. *Infrastructures*: • Trinity College Dublin and the ADAPT Centre have established the **AI Accountability Lab**, dedicated to exploring issues of fairness, transparency, and accountability in AI. • **Data sharing initiatives**: National Energy Research Database; National Open Research Forum; Ireland's Open Data Portal; Surveying Open Data Portal; Geospatial Data Hub. • The Higher Education Authority funded key initiatives in 2022 to advance Open Research, culminating in the establishment of the **national data stewardship network**, Sonraí. *Talent*: • Research Ireland's Centre for Research Training in Artificial Intelligence aims to develop skilled researchers in cutting-edge AI technologies.
Italy	*Policy framework*: • The national project on AI, named **FAIR (Future AI Research)**, funded with Next Generation EU funds, has a transversal project on AI4Science&Science4AI. It has been a bottom-up initiative, and there is no specific funding programme on this. • Italy has recently established **AI4Industry**, an institute focusing on automotive, aerospace and AI-research. *Talent*: • Italy's **National PhD Programme in AI** provides an interdisciplinary AI curriculum, training researchers in AI for scientific discovery, health, and industry applications.

Member State/ Associated Country	Relevant examples of programmes for AI in Science
	• Italy supports **international mobility** through Industrial PhD Scholarships and significant tax breaks for researchers returning from abroad.
The Netherlands	*Policy framework*: • In 2019, the Netherlands Organisation for Scientific Research (NWO) published an **AI research agenda** that set out grand challenges relating to both AI capabilities and the application of AI in different domains. • The National Growth Fund (NGF) funds AiNed with 189 million EUR. This programme includes a subprogramme on Knowledge and Innovation, which in part enables AI for research and discovery. • **Interdisciplinary research** is promoted via initiatives like the Innovation Centre for AI (ICAI), and e.g. interdisciplinary labs at e.g., TU Delft, TU Eindhoven's cross-departmental AI institute (EAISI) and the University of Amsterdam's AI4Science Lab. *Infrastructures*: • **SURF.NL 2024** Call for consultancy and computational resources for the development of a scientific use case in machine learning-assisted data analysis, modelling, and interpretation. *Talent*: • Research funding initiatives that **promote mobility**, such as the ICAI, AiNed XS Europe (for international collaboration), and attractive research funding programmes by NWO. • **NGF AiNed Fellowship Grant programme**, which seeks to draw AI-talent to Dutch academic research institutions, recognising the global competition for AI expertise. • **ICAI** brings universities and industry together to offer **specialised AI-training** for experts in various fields, helping bridge the gap between research and real-world applications.
Norway	*Policy framework*:

Member State/ Associated Country	Relevant examples of programmes for AI in Science
	• The Government has recently published a **white paper** on the Norwegian research system where the impact of AI on research is a major topic. • The Government has a **strategy for digitalisation in the public sector** with the objective that 80% of all public sector organisations are to implement AI in their operations by the end of 2025. • The RCN AI-research funding programme actively promotes AI-industry collaboration by offering joint funding opportunities, ensuring that researchers in both sectors benefit from shared knowledge and resources. One of the objectives of the programme is to promote the use of AI in Norwegian research institutions. *Infrastructures*: • The Norwegian AI Cloud (NAIC) has been established to help researchers use AI in their research.
Romania	*Policy framework*: • A new **AI Hub** was launched in 2025 with EU funding, at the Politechnica University in Bucharest with antennas in Romanian regions. It is to be determined how much it will also focus on AI in science and AI interdisciplinary research. *Infrastructures*: • Medicai is a **cloud-based medical imaging platform** which is used by multiple hospitals and clinics across Romania. Medicai allows for the upload, sharing, and analysis of medical images (CT, MRI, X-rays) in a secure and efficient manner.
Slovenia	*Policy framework*: • In 2024, Slovenia introduced **strategic priority projects - Gravity** (2.25 million EUR per project), two of which focus on the AI domain (LLMs for digital Humanities and AI for Science). *Infrastructures*:

Member State/ Associated Country	Relevant examples of programmes for AI in Science
	• **EuroHPC Vega** is running from 2021, and in February 2025 the application for an AI Factory with Vega 2 HPC was confirmed (expected to be operational in 2027). • **Open Science Slovenia** - a national Open Science Portal and repositories for open access to theses and research results of researchers at the Slovenian universities. • The **Spoznaj project**, which provides comprehensive training for data stewards, focusing on the implementation of open science. • In 2024, the career position of **data steward** was introduced through the revision of legislation for the public sector. • The **Oncochain platform**, a private initiative, facilitates the collection, curation, and sharing of oncology data from hospitals and clinics. *Talent*: • Mobility is supported by the implementation of funding for reintegration projects open to **MSCA fellows** and by the funding of the **MSCA Seal of Excellence** extended by the reintegration project (RRF funds). In addition, there is the reintegration programme for early career Slovenian researchers abroad through the "**Dr. Aleš Debeljak**" call for projects (national funds). All mobility schemes are open to all scientific fields. • MSCA Co-fund project **SMASH (Machine Learning for Science and Humanities Postdoctoral Programme)** coordinated by the University of Nova Gorica is performing post-doctoral training on cutting-edge ML applications and AI techniques in selected scientific fields.
UK	*Policy framework*: • UK Research and Innovation (UKRI) has invested £100 million in a network of nine **AI for science hubs**, each with different technical and domain specialisms. • The **Accelerate Programme for Scientific Discovery**, based at the University of Cambridge, aims to advance scientific research by integrating AI across disciplines. Supported by a donation from Schmidt Sciences, Accelerate Science was established in 2020 and supports three core pillars of activity: research, training, and community-building.

Member State/ Associated Country	Relevant examples of programmes for AI in Science
	• In 2023, the previous UK Government announced a £100 million **AI for health mission** that aims to accelerate the adoption of AI in life sciences. • The UK has invested £155 million in a 10-year programme to **digitise natural science collections in UK museums**. The creation of such rich data resources could provide materials to support a range of novel AI for science applications. *Infrastructures*: • **Polymathic AI** is a research collaboration that brings together experts at the Simons Foundation, New York University, University of Cambridge, Princeton University, and the Lawrence Berkeley National Laboratory. Its goal is to accelerate the development of versatile foundation models. *Talent*: • A 2022 **Fellowship scheme** by UKRI provided £4.5 million of funding to support up to eight Research Software Engineering fellowships at UK research institutions. • The **Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship schemes based** at the University of Oxford and Imperial College London support early career researchers to pursue innovative AI applications in STEM research. • The Alan Turing Institute in the UK focuses on AI and data science research, **promoting partnerships** between computer scientists, statisticians, and domain experts from fields like medicine and climate science.

Table 2: Relevant examples of programmes for AI in Scienc

Annex 2: Methodology

The MLE on National Policies for AI in Science followed the standard methodology for conducting MLEs within the framework of the Horizon Europe Policy Support Facility, drawing on the approach outlined in "*Mutual Learning Exercise - a new methodology*".[37] As a Member-State and Associated Country-driven and policy challenge-based initiative, the MLE promotes mutual learning among participating countries by facilitating the exchange of experiences, challenges, and best practices.

Over the course of a year (July 2024-June 2025), the participating countries engaged in a structured dialogue to explore the best approaches to addressing policy challenges related to AI in science. The exercise provided a platform for policy makers and experts to compare national policies, analyse implementation strategies, and assess the short- and long-term impact of AI adoption in scientific research. A key aim of the MLE was to support evidence-based policymaking, helping countries optimise the design and implementation of AI in science related policies while learning from one another's experiences.

The MLE methodology was based on open, frank, and confidential knowledge exchange among participants. All countries were expected to actively contribute, provide relevant data and information, and share their policy approaches and challenges. While each country had specific knowledge and interests, the issues addressed were sufficiently common to ensure mutual learning and broad relevance to identify effective AI policy frameworks in science.

Distribution of work

1. **Participating countries**. Each participating country appointed high-level representatives with expertise in R&I policy. These representatives contributed to the MLE by providing national data, sharing policy experiences, and engaging in discussions. They also allocated time for attending meetings, contributing to thematic reports, and participating in country visits.
2. **Independent experts**. The MLE was supported by a team of independent experts selected by the European Commission, including:
    - Jessica Montgomery (Chair and Expert on Topic 1: Framework conditions and funding)
    - Stefano Bianchini (Rapporteur)
    - Ross D. King (Expert on Topic 2: Infrastructures for AI in science)
    - Vida Groznik (Expert on Topic 3: Talent for AI in science)

    These experts were responsible for preparing background analyses, drafting discussion papers and thematic reports, shaping the agenda for their respective MLE topic meetings, and synthesising the main insights from the MLE discussions. The rapporteur contributed to the final report, summarising the key learnings from the MLE process.
3. **European Commission services.** The Directorate-General for Research and Innovation (DG RTD) actively supported the MLE by providing context on the EU-level policy framework and outlining ongoing policy challenges and relevant actions. Units A.1 (Semester & Country Intelligence) and E.4 (Industry 5.0 & AI in Science) were involved in the MLE. The main contacts from the European Commission were Laura Roman, Annamaria Zonno, Daniela Petkova, and Vasileios Frantzis.

[37] European Commission: Directorate-General for Research and Innovation and Luukkonen, T., *Mutual Learning Exercises – A proposal for a new methodology* – Horizon 2020 Policy Support Facility, Publications Office of the European Union, 2016, https://data.europa.eu/doi/10.2777/292023

4. **PSF contractor.** Within the PSF consortium, Ecorys was responsible for organising meetings and country visits, compiling meeting minutes and ensuring the overall development of the MLE, and publishing materials. The project manager was Gillian Kelly, and the quality control was provided by Alasdair Reid of EFIS Centre.

Sources of information for the final report

This report builds on five complementary sources of information:

- **Thematic reports by independent experts.** The findings of three thematic reports prepared by Jessica Montgomery, Ross D. King, and Vida Groznik. Elements of these reports have been reproduced or adapted for inclusion in the final report. All three thematic reports are available for downloaded on the MLE website here.
- **Background literature and policy documents.** The thematic reports include reviews of relevant scholarly literature and policy documents, outlining current and past activities undertaken at the European level and within countries to support AI in science. To provide further context for the MLE, additional desk research was conducted for this final report.
- **Presentations on national or institutional initiatives.** Throughout the MLE meetings and country visits to Italy, Norway, and Belgium, invited speakers presented specific national and institutional initiatives related to AI in science.
- **Discussions among MLE participants**. Exchanges among MLE participants during country visits and online meetings. These were documented in meeting notes, which were compiled and organised by Gillian Kelly.
- **Survey responses.** A structured questionnaire was sent to country representatives and chief scientists on each of the three respective topics of the MLE to collect specific examples and perspectives on the current state of AI policies in science across participating countries. The responses were then analysed by the thematic experts and included in the respective topic discussion papers and thematic reports.

Annex 3: List of country representatives who participated in the MLE

Country	Participant	Affiliation
Austria	**Barbara Buzeczki**	Federal Ministry for Education, Science and Research, Austria
Austria	**Georg Sedlbauer**	Federal Ministry for Climate Action, Environment, Energy, Mobility, Innovation and Technology, Austria
Austria	**Stefan Hanslik**	Federal Ministry for Education, Science and Research, Austria
Belgium (Flanders)	**Peter Spyns**	Flemish Public Administration- Department of Work, Economy, Science, Innovation & Social Economy, Belgium
Estonia	**Aile Tamm**	Ministry of Education and Research of Estonia
Germany	**Florentin Neumann**	Deutsche Forschungsgemeinschaft (DFG), German Research Foundation
Germany	**Michael Schmuker**	Helmholtz Association, Germany
Germany	**Christian Beilmann**	Helmholtz Information and Data Science Academy, Germany
Hungary	**Adam Ónodi-Szabó**	Department of Science Policy and Research Infrastructures, Ministry of Culture and Technology, Hungary
Ireland	**Kerrie Sheehan**	Sustainable Energy Authority of Ireland
Ireland	**Irene Ward**	Sustainable Energy Authority of Ireland
Ireland	**Neetu**	Research Ireland
Italy	**Fosca Giannotti**	Scuola Normale Superiore di Pisa, Italy
Italy	**Roberto Cimino**	Ministry of University and Research, Italy

Country	Participant	Affiliation
Netherlands	**Jennifer Lieuw**	Ministry of Education, Culture and Science, Netherlands
Netherlands	**Anthony Gadsdon**	Dutch Research Council, Netherlands
Netherlands	**Marieke Van Duin**	Dutch Research Council, Netherlands
Netherlands	**Linda Lieferink**	Dutch Research Council, Netherlands
Norway	**Philip Lorentzen**	Research Council of Norway
Romania	**Raluca Ciobotaru**	UEFISCDI, Research Funding Agency, Romania
Romania	**Vlad Enache**	UEFISCDI, Research Funding Agency, Romania
Slovenia	**Andreja Umek Venturini**	Ministry of Higher Education, Science and Innovation, Slovenia
Slovenia	**Maruša Arh**	Ministry of Higher Education, Science and Innovation, Slovenia

Table 3: List of country representatives who participated in the MLE

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Artificial Intelligence (AI) promises to accelerate the pace of scientific discovery and revolutionise how we uncover and use new knowledge. Future AI systems will become ever more powerful and accessible. To stay competitive, Europe must act: AI in science needs strategic, coordinated policies. Eleven countries joined the Mutual Learning Exercise (MLE) to share experiences and identify key levers across funding, infrastructure, and talent. The MLE highlighted the need for adaptive, long-term policy frameworks that combine top-down investment with bottom-up experimentation. Human and technical infrastructure, ethical governance, and open, FAIR data are needed to enable responsible, widespread adoption of AI in scientific research. The resulting recommendations offer a flexible policy menu tailored to different stages of AI maturity in science.

Studies and reports

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